Update to the Leps Strength Envelopes for Rockfill Based on 900 Large-Scale Triaxial Tests

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Abstract: This study presents a database of over 900 large-scale triaxial tests that were used to calibrate and refine the Leps strength envelopes for rockfill materials. A novel nonlinear shear strength criterion for rockfill is introduced that can replicate the Leps strength envelopes, is derived in terms of both shear/normal and principal stresses, employs a single fitting parameter with direct physical significance, and allows for smooth transitions between rockfill strength categories. An accompanying empirical model is presented to estimate rockfill strength based on void ratio and rock type, offering a practical tool for rapid application. The model simplifies to five strength envelopes and demonstrates strong predictive performance, while maintaining simplicity. The evaluation of prediction accuracy indicates that the strength model is applicable for soils with a median particle size greater than 2 mm and a fines content of less than 25%. Analysis by stress range reveals a limitation of power-law strength criteria: when calibrated using tests at very low confining stresses, power-law models can significantly over- or underestimate strength for normal stresses above 100 kPa. To address this, a 60-600 Guideline is proposed, recommending the exclusion of triaxial confining stresses below 60 kPa and inclusion of at least one test with a confining stress above 600 kPa to obtain reliable calibration of power-law strength criteria for rockfill from laboratory testing data. An improved methodology to convert triaxial test results into shear and normal stresses is also introduced, accounting for the nonlinearity of the failure envelope.
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Introduction

Rockfill is an important construction material extensively used in civil engineering projects, including dams, embankments, railway ballast, and erosion protection (Leps 1988; Indraratna et al. 1998). In the mining industry, large rockfill slopes reaching heights of several hundred meters are often constructed in stages to store waste rock and have the potential to pose significant challenges for stability, safety, and environmental management if not designed properly (Valenzuela et al. 2008; Linero Molina et al. 2019). The geotechnical stability of these structures depends on the shear strength of rockfill, a property governed by many factors, including the confining stress level, particle size distribution (Ovalle et al. 2014; Cantor et al. 2024; Carrasco et al. 2025), particle shape (Ovalle and Dano 2020; Carrasco et al. 2023; Quiroz-Rojo et al. 2025; Aponte et al. 2025), particle strength (Alonso et al. 2012; Tapias et al. 2015; Yan et al. 2022), particle breakage (Jia et al. 2017; Jin et al. 2024; Xiao et al. 2014b), and degree of compaction (Marsal 1973; Indraratna et al. 1993). In the last decade, field-scale testing (Girumugisha et al. 2024) and numerical modeling (Azéma et al. 2017, 2021; Bignon et al. 2023; Binaree et al. 2020, 2021, 2024; Polanía et al. 2023, 2024) have further refined the understanding of the factors that affect rockfill shear strength.

Rockfill sourced by screening fluvial sediments typically consists of rounded particles with sizes up to 40 mm, whereas rockfill produced by blasting activity may contain more angular particles with maximum sizes often exceeding 1,000 mm (Deiminiat et al. 2020). Testing the strength of such large granular materials is inherently challenging, because standard laboratory triaxial tests are often limited to smaller particles that may not produce results representative of field conditions. To address these limitations, researchers over the last five decades have developed and refined rockfill shear strength criteria using published data from a limited number of large-scale tests.

Several notable shear strength models for rockfill have been developed utilizing power-law or logarithmic formulations to account for the nonlinearity of the failure envelope (de Mello 1977; Charles and Watts 1980; Barton and Kjærnsli 1981; Gonzalez Garcia 1985; Charles 1991; Indraratna et al. 1993). Many models have incorporated particle breakage mechanics and demonstrated how grain shape, angularity, and interlocking influence shear strength (Frossard et al. 2012; Ovalle et al. 2014; Rahmani and Panah 2021; Carrasco et al. 2022). Fractal-based models considering the effects of crushing on the particle size distribution have also been introduced (Xu et al. 2015). Although these models offer valuable insights into the mechanics of rockfill behavior, their practical application is often constrained by the need for calibration to laboratory data.

In 1970, Leps analyzed data from 109 large-scale triaxial tests and categorized the results into three nonlinear strength envelopes, recognizing that the secant friction angle decreases with increasing

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normal stress on the failure plane. These envelopes have been used to estimate rockfill shear strength when site-specific testing data are unavailable and have been widely applied in civil and mining projects worldwide (Hawley and Cunning 2017). However, the Leps criteria lack a consistent, quantitative method for determining which of the three strength categories should be applied to a specific material.

This study addresses these challenges by compiling a database of over 900 published large-scale strength tests on rockfill that is used to validate and enhance the power-law strength envelopes proposed by Leps. The key contributions of this work include the proposal of a new failure criterion that (1) reproduces and smoothly transitions between the quantitative strength categories, (2) contains a single fitting parameter with direct physical meaning and clear accounting of units, and (3) can be interpreted in terms of both shear/normal and principal stresses. Additionally, a new empirical model is developed to guide practitioners in selecting appropriate strength envelopes for rockfill based on the field-measurable properties of compaction and rock type. The accuracy of this model and failure criterion are evaluated by stress range to provide practical guidance for laboratory testing of rockfill. Finally, a new definition is proposed to specify the range of soil types that exhibit the mechanical behavior of rockfill, based on the demonstrated applicability of the proposed model across a diverse range of rockfill sample characteristics.

Failure Criterion

For soil under a condition of direct shear, the failure criterion is stated in a power-law form as

$$\tau = K p_a \left(\frac{\sigma_n}{p_a}\right)^m \tag{1}$$

where τ and σ_n = shear and effective normal stresses on the failure plane, respectively; K = strength coefficient; p_a = standard atmospheric pressure; and m = curvature coefficient. The K parameter controls the inclination of the failure envelope and the m parameter controls the curvature of the envelope. The introduction of p_a normalizes the power-law criterion so that the parameter K is unitless and invariant for any system of units, and is defined as a constant: $p_a = 0.1013$ MPa $= 101.3$ kPa $= 14.70$ psi $= 2,116$ psf.

The simple power-law criterion is formulated in shear/normal stress space and therefore the curvature coefficient m is restricted between 0.5 and 1.0 (Yu et al. 2020; Anyaegbunam 2015). When $m = 0.5$, the failure envelope has the maximum curvature possible while still adhering to Mohr failure laws. When $m = 1.0$, the failure envelope reduces to a linear Mohr–Coulomb relationship. A power-law envelope for rockfill was first proposed by de Mello (1977), and later, Gonzalez Garcia (1985) developed a dimensionless version similar to Eq. (1).

Differentiating the criterion with respect to σ_n results in the tangent of the instantaneous friction angle (φ) as shown in the following equation:

$$\tan\varphi = \frac{d\tau}{d\sigma_n} = Km\left(\frac{\sigma_n}{p_a}\right)^{m-1} \tag{2}$$

Combining Eqs. (1) and (2) leads to a formulation of the criterion expressed in terms of tan φ

$$\tau = \sigma_n \frac{\tan\varphi}{m} \tag{3}$$

The criterion is given physical significance by defining K in terms of the secant friction angle ($\varnothing'_S$), which is calculated with Eq. (4) and derived from the linear Mohr–Coulomb relation by setting the cohesion to zero and solving for the friction angle

$$\varnothing'_S = \operatorname{atan}\left(\frac{\tau}{\sigma_n}\right) \tag{4}$$

The tangents of the instantaneous and secant friction angles are therefore related by the parameter m

$$\tan\varphi = m\tan\varnothing'_S \tag{5}$$

By setting σ_n to $10p_a$ and τ to $10p_a \cdot \tan\varnothing'_1$ in Eq. (1), the parameter K can be expressed as follows:

$$K = \tan\varnothing'_1 10^{1-m} \tag{6}$$

where $\varnothing'_1$ = secant friction angle at a normal stress of $10p_a$ (approximately 1 MPa). This stress level is referred to as the anchor stress because, at this point, the strength is not dependent on the exponent m. While the choice of $10p_a$ as the anchor stress is somewhat arbitrary, it was selected for two reasons: (1) it is a stress level commonly encountered in both laboratory and field conditions, and (2) this anchor stress resulted in improved correlation between $\varnothing'_1$ and key test parameters, such as void ratio. Integrating Eq. (6) into Eq. (1) leads to a formulation of the strength criterion with $\varnothing'_1$ and m as the fitting parameters as shown in Eq. (7)

$$\tau = \tan\varnothing'_1 10^{1-m} p_a \left(\frac{\sigma_n}{p_a}\right)^m \tag{7}$$

The criterion may be simplified as shown in the following equation:

$$\tau = \tan\varnothing'_1 (10p_a)^{1-m} \sigma_n^m \tag{8}$$

A Mohr diagram of the criterion showing the stresses at failure for the condition of $\sigma_n = 10p_a$ is presented in Fig. 1. Based on this study and previous research (Douglas 2002; Cylwik et al. 2023), a typical value for the curvature coefficient of rockfill is $m = 0.9$. Taking a value of 0.9 for m, the proposed failure criterion for rockfill is obtained in Eq. (9) with $\varnothing'_1$ as the only fitting parameter

$$\tau = \tan\varnothing'_1 (10p_a)^{0.1} \sigma_n^{0.9} \tag{9}$$

Eq. (9) replicates the three rockfill shear strength envelopes introduced by Leps (1970) when $\varnothing'_1$ is equal to 36° (low density/poorly graded/weak rockfill), 41° (average rockfill), or 46° (high density/well graded/strong rockfill), as shown in Fig. 2.

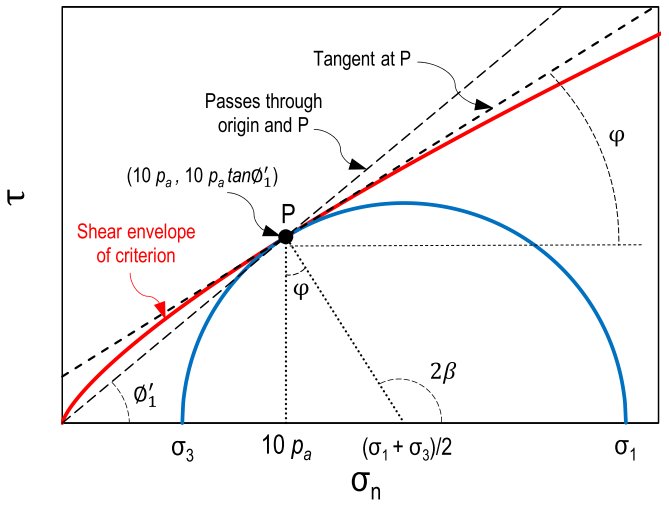


Fig. 1. (Color) Mohr diagram of the proposed failure criterion.

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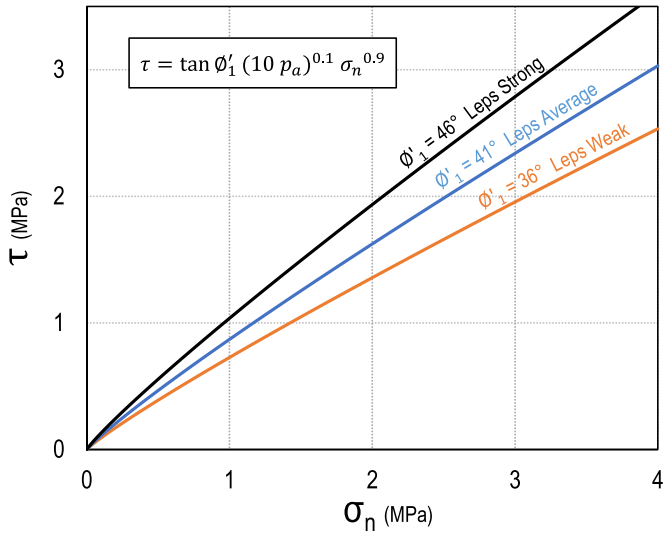


Fig. 2. (Color) Shear strength envelopes with the proposed criterion that replicate the Leps (1970) strength envelopes for rockfill materials.

Table 1. Common shear strength envelopes for rockfill that are replicated by the proposed criterion ($m = 0.9$)

$\emptyset'_1$ (degrees)	Leps (1970) rockfill	Hawley and Cunning (2017) rockfill	Indraratna et al. (1993) rockfill
33	—	—	Lower limit strength
36	Weak/loose	Top-down construction Very high to high hazard	—
38	—	Top-down construction Moderate-to-low hazard	—
41	Average	—	—
43	—	Bottom-up construction Very high to high hazard	—
46	Strong/ dense	Bottom-up construction Moderate-to-low hazard	—

Additionally, it closely reproduces the strength envelopes proposed by the Large Open Pit Project on mine waste dump and stockpile design (Hawley and Cunning 2017) and the lower bound rockfill shear strength envelope proposed by Indraratna et al. (1993), as summarized in Table 1.

Derivation in Principal Stresses

Stress pairs at failure in σ_n and τ may be converted to the major (σ_1) and minor (σ_3) effective principal stresses at failure using equations rearranged from Balmer (1952) as shown in Eqs. (10)–(12) (Cylwik et al. 2024), where β is the predicted angle of the failure plane with respect to σ_3

$$\sigma_1 = \sigma_n + \tau \cdot \tan\beta \tag{10}$$

$$\sigma_3 = \sigma_n - \tau \cdot \frac{1}{\tan\beta} \tag{11}$$

$$\beta = \frac{\varphi}{2} + \frac{\pi}{4} \; = \frac{1}{2}\text{atan}\left(\frac{d\tau}{d\sigma_n}\right) + \frac{\pi}{4} = \frac{1}{2}\text{atan}\left(Km\left(\frac{\sigma_n}{p_a}\right)^{m-1}\right) + \frac{\pi}{4} \tag{12}$$

According to Mohr failure theory, σ_n and τ can be defined as

$$\sigma_n = \frac{\sigma_1 + \sigma_3}{2} - \frac{\sigma_1 - \sigma_3}{2}\sin\varphi \tag{13}$$

$$\tau = \frac{\sigma_1 - \sigma_3}{2}\cos\varphi \tag{14}$$

Combining Eqs. (3), (13), and (14) leads to a quadratic equation with an unknown sinφ

$$\left(\frac{\sigma_1 - \sigma_3}{2} - m\frac{\sigma_1 - \sigma_3}{2}\right)\sin^2\varphi - \frac{\sigma_1 + \sigma_3}{2}\sin\varphi + m\frac{\sigma_1 - \sigma_3}{2} = 0 \tag{15}$$

Taking the positive root of Eq. (15) for sinφ results in

$$\sin\varphi = \frac{\sqrt{4m(m-1)(\sigma_1 - \sigma_3)^2 + (\sigma_1 + \sigma_3)^2} - (\sigma_1 + \sigma_3)}{2(m-1)(\sigma_1 - \sigma_3)} \tag{16}$$

Eqs. (1), (6), (13), (14), and (16) can be combined to define $\emptyset'_1$ in terms of σ_1 and σ_3 at failure

$$\tan\emptyset'_1 = (20p_a)^{m-1}(\sigma_1 - \sigma_3)\sqrt{1 - \sin^2\varphi}(\sigma_1 + \sigma_3 - (\sigma_1 - \sigma_3)\sin\varphi)^{-m} \tag{17}$$

Eq. (15) can be rearranged to the form of a traditional failure criterion in terms of σ_1 and σ_3

$$\sigma_1 = \sigma_3 + \sigma_3\frac{2\sin\varphi}{(1-m)\sin^2\varphi - \sin\varphi + m} \tag{18}$$

Eq. (18) can also be expressed in terms of $\tan^2\beta$

$$\sigma_1 = \sigma_3 + \sigma_3\frac{\tan^4\beta - 1}{\tan^2\beta\,(2m-1) + 1} \tag{19}$$

$$\tan^2\beta = \frac{d\sigma_1}{d\sigma_3} = \tan^2\left(\frac{\varphi}{2} + \frac{\pi}{4}\right) = \frac{1 + \sin\varphi}{1 - \sin\varphi} \tag{20}$$

By combining Eqs. (2), (3), (14), and (18), σ_3 is isolated in terms of φ

$$\sigma_3 = p_a\frac{\sqrt[1-m]{Km\cot\varphi}\;(\text{m}(\sin\varphi + 1) - \sin\varphi)}{m\;(\sin\varphi + 1)} \tag{21}$$

Eq. (21) is used to iteratively calculate φ for a given σ_3, and Eq. (18) is then applied to calculate the corresponding σ_1 at failure. Because of the presence of the irrational root in Eq. (21), isolating φ explicitly in terms of σ_3 is not currently possible. However, for the special case of $m = 0.5$, Eq. (21) can be solved for φ. Integrating this solution into Eq. (18) and simplifying yields the failure criterion in terms of the principal stresses for $m = 0.5$ and $\sigma_3 > 0$. Eq. (22) is verified by the derivation presented in Anyaegbunam (2015) for the generalized case of a power-law criterion with an intercept

$$\text{For } m = 0.5, \;\; \sigma_1 = \left(K\sqrt{p_a} + \sqrt{\sigma_3}\right)^2 \tag{22}$$

For projects requiring an explicit closed-form solution in terms of principal stresses, an approximation for φ as a function of σ_3 has been developed, as shown in Eq. (23). When combined with Eq. (18), σ_1 at failure is approximated with an error of less than 1% for the conditions of $0.7 < m < 1$ and $25° < \emptyset'_1 < 50°$

$$\tan\varphi \approx (K(1.714\,\text{m} - 0.714) - 0.289\ln(\text{m}) - 0.007) \times \left(\frac{\sigma_3}{p_a}\right)^{(0.528\text{m}^2 + 0.0021\text{m} - 0.528)} \tag{23}$$

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Laboratory Testing Database

To analyze and validate the proposed strength criterion, a database of 934 large-scale triaxial tests conducted on rockfill with measurement of effective stress parameters was used, in addition to four direct shear tests. The data set builds on the compilation by Douglas (2002) completed as part of his thesis at the University of New South Wales. All data were sourced from published research, encompassing 41 publications and 31 distinct rock types. While testing procedures varied across studies, triaxial tests were typically conducted under consolidated drained (CD) conditions, with measurements of particle size distribution, density, and void ratio reported for each test. Prior to testing, the samples were screened to remove oversized particles according to local testing procedures.

Notable large-scale triaxial devices in the database include a 1,130 mm-diameter triaxial cell by Marsal et al. (1965), a 1,000 mm-diameter cell by Hu et al. (2011), and a 914 mm cell by Marachi et al. (1972). Although most tests were included in the database, a small subset was excluded. Tests were discarded if they lacked key data, such as rock type or initial void ratio, because these parameters were used as direct inputs for the proposed empirical model. Additionally, tests resulting in $\varnothing'_1 < 31°$ were excluded, because it was reasoned that these likely reflect shear behavior influenced by fine-grained soils rather than rockfill. The database has been made available by the authors in the "Data Availability Statement" section.

Table 2 provides a summary of the univariate statistics for select parameters in the database, illustrating the wide variety of rockfill types included. The tested samples span a broad range of particle sizes, gradation types, and compaction levels. Triaxial cell diameters ranged in size from 51 to 1,130 mm, with an average diameter of 396 mm. Maximum particle sizes of samples tested varied from 4.8 to 200 mm. On average, samples contained 78% gravel content, with little to no fine-grained soil. Only 10% of the tests had a fines content greater than 20%. The uniaxial compressive strength (UCS) of the particles exhibited a large range of 25–300 MPa; the UCS is typically measured from cored samples of the rockfill material following procedures in ASTM D7012-14e1 (ASTM 2023a). To standardize particle angularity across the various classification systems, each sample was assigned an angularity rating on a scale of 1 (rounded) to 8 (angular). Notably, 76% of the samples received the highest angularity rating of 8, suggesting that the qualitative assessments reported in the literature may lack the sensitivity needed to fully capture the underlying variability of particle angularity.

Conversion to Shear and Normal Stresses

Most published research on the strength of rockfill has estimated σ_n and τ stress pairs from triaxial test results, typically measured at failure in terms of σ_1 and σ_3, using what is referred to herein as the linear approximation method. This method assumes that the secant friction angle defined in terms of principal stresses, also known as the peak-state friction angle ($\varnothing'_{PS}$), is equal to φ for the tested value of σ_3. Several equivalent forms to calculate $\varnothing'_{PS}$ are summarized in Eq. (24) (Charles and Soares 1984; Cylwik et al. 2024). However, Eq. (5) demonstrates that for a nonlinear failure envelope (i.e., $m < 1$), the instantaneous friction angle is always less than the secant friction angle (Baker 2004). The assumption that $\varnothing'_{PS} = \varphi$ is valid only when the failure envelope is linear (i.e., $m = 1$). Laboratory tests have consistently shown that the shear envelope of rockfill is nonlinear (Indraratna et al. 1998), and therefore, the linear approximation method introduces errors when converting rockfill triaxial test results to σ_n and τ. Fig. 3 illustrates a Mohr diagram showing that $\varnothing'_{PS} \neq \varphi$ when the failure envelope is nonlinear

$$\varnothing'_{PS} = \operatorname{asin}\left(\frac{\sigma_1 - \sigma_3}{\sigma_1 + \sigma_3}\right) = \operatorname{asin}\left(\frac{(\sigma_1/\sigma_3) - 1}{(\sigma_1/\sigma_3) + 1}\right) = 2\operatorname{atan}\left(\sqrt{\frac{\sigma_1}{\sigma_3}}\right) - \frac{\pi}{2} = \operatorname{asin}\left(\frac{\tau}{\tau \sin\varphi + \sigma_n \cos\varphi}\right) \quad (24)$$

Example stress pairs calculated using Eqs. (8), (10), and (11) for m values ranging from 0.5 to 1 are shown in Table 3, and demonstrate a systematic underestimation of σ_n and τ when the linear approximation method is applied to convert stress pairs from a nonlinear envelope. The error increases for more curved shear envelopes (lower m values) and diminishes as the failure envelope approaches linearity ($m = 1$). For most samples in the database, the linear approximation method introduces minimal error since $m > 0.8$ is typical for rockfill. However, in 150 tests, the linear approximation method produces an error exceeding 5%, with an average error of 13% for those tests.

To account for the curvature of the failure envelope, the following iterative procedure was employed to convert the triaxial test results to σ_n and τ stress pairs:

1. Estimate the parameter m for a triaxial testing set by first selecting trial values of m and $\varnothing'_1$. A test set is a group of tests performed on the same material at different confining pressures. For each test in the set, trial values were calculated for φ using Eq. (16), σ_n using Eq. (13), τ using Eq. (14), and σ_1 using Eq. (10). The Generalized Reduced Gradient (GRG) nonlinear solver (available with the Microsoft Excel Solver function) was utilized to find the best estimates of m and $\varnothing'_1$ to match the trial σ_1 values to the tested values. An alternative option to estimate m and $\varnothing'_1$ would be to use Eqs. (10) and (11) to generate trial failure envelopes in $\sigma_1 - \sigma_3$ stress space and perform a traditional regression to the triaxial test results.
2. Calculate φ for each test using Eq. (16), the triaxial test result, and the estimated value of m for the set.
3. Calculate σ_n and τ for each test using Eqs. (13) and (14), the triaxial test result, and the estimated value of φ for the test.

Table 2. Univariate statistics of select sample parameters from the database (938 tests)

Parameter	Symbol	Minimum	Mean	Maximum	Median	Std. Dev.
Test diameter (mm)	D_s	50.8	396	1,130	229	353
Median particle size (mm)	D_{50}	0.21	18	100	8.5	21
Maximum particle size (mm)	D_{max}	4.8	67	200	38	60
Gravel content (%)	%Gravel	22	78	100	84	21
Fines content (%)	%Fines	0	3.2	45	0.50	6.8
Coefficient of uniformity	C_u	1.4	42	1,714	9.0	158
Coefficient of curvature	C_c	0.04	2.0	17	1.4	2.1
Void ratio at start of test	e	0.15	0.43	0.89	0.41	0.14
Uniaxial compressive strength (MPa)	UCS	25.0	159	300	200	65.0
Particle angularity rating	Ang	2.0	7.0	8.0	7.1	1.3

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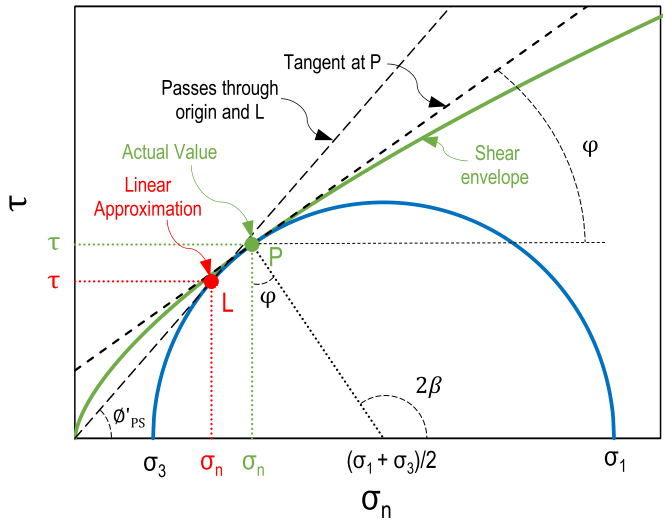


Fig. 3. (Color) Mohr diagram comparing the linear approximation method with the actual values of σ_n and τ for a nonlinear failure envelope.

Shear Strength

The complete test data are shown in Fig. 4 in terms of σ_n and τ. Linear regression through the database yielded a slope of 0.80, equivalent to a friction angle of 38.7°, with a coefficient of determination (R^2) value of 0.981. When the power-law criterion in Eq. (8) was applied, the R^2 increased to 0.982, indicating only a marginal improvement over the linear fit. Fig. 5 shows σ_n versus the measured $\varnothing'_S$ for each test in the database, alongside five different strength envelopes calculated using the proposed criterion in Eq. (9). This chart format mirrors the one originally used by Leps (1970) to evaluate the strength of rockfill. The data exhibit a clear trend of decreasing $\varnothing'_S$ as σ_n increases, a phenomenon well documented in the literature (Charles and Watts 1980; Xiao et al. 2016; Ovalle et al. 2020).

For each test set, a best-fit regression was performed by minimizing the sum of the squared error using the generalized shear strength model in Eq. (8). Fig. 6 illustrates two example test sets in the database fitted with Eq. (8). The database contains 249 test sets, each consisting of a series of triaxial or direct shear tests. Histograms of the resulting $\varnothing'_1$ and m values are shown in Figs. 7(a and b), respectively. The mean value of $\varnothing'_1$ is 41.3° and is consistent with the average shear strength envelope predicted by Leps (1970).

The correlation coefficient (R) between $\varnothing'_1$ and m for all test sets is $R = -0.12$, demonstrating the relative independence of the two fitting parameters and indicating a weak correlation between shear strength and curvature of the failure envelope. The correlation coefficient helps reveal how strongly each sample parameter influences the test results, and ranges from –1 (perfect negative correlation) to 1 (perfect positive correlation). In contrast, selecting Eq. (1) as the failure criterion would result in a stronger correlation coefficient of $R = -0.80$ between K and m, which is consistent with the findings of Muñiz-Menéndez and Estaire (2021) for that type of criterion. The high correlation between K and m suggests significant interdependence. In comparison, the independence of the fitting parameters in Eq. (8) provides an advantage, enabling the strength ($\varnothing'_1$) and curvature of the failure envelope (m) to be analyzed separately without blending their effects.

Empirical Strength Model

Development of Model

The correlation coefficient was calculated between the stress ratio at failure, $(\tau/\sigma_n)_f$, and various sample characteristics for each test in the database to evaluate their influence on shear strength (Table 4). The sample parameters considered included the initial void ratio (e), gravel content (%Gravel), median particle size (D_{50}), coefficient of uniformity (C_c), coefficient of curvature (C_c), UCS, and particle angularity rating (Ang). Eqs. (3) and (4) demonstrate that $(\tau/\sigma_n)_f$ is directly related to both φ and $\varnothing'_S$. Table 4 shows that, on average, none of the parameters exhibited a strong correlation with $(\tau/\sigma_n)_f$ on a test-by-test basis. While previous studies (Barton and Kjærnsli 1981) have suggested that these parameters can influence rockfill shear strength, their effects might be too subtle to be observable above the noise in the data set.

To reduce noise and normalize the measured strength values for different stress levels, the database was analyzed by treating each test set as an individual data point by considering the fitting parameters $\varnothing'_1$ and m. The strongest correlation observed was between $\varnothing'_1$ and e, with a correlation coefficient of $R = -0.34$. Fig. 8 presents a scatterplot of this relationship and the best-fit linear regression with $R^2 = 0.114$ (note that the coefficient of determination for a linear regression is the square of the correlation coefficient, R). An inverse relationship between void ratio and rockfill strength has also been demonstrated by other researchers (Leps 1970; Indraratna et al. 1993).

Further analysis of the database revealed a relationship between rock type and shear strength. For example, Fig. 9 compares the results of test sets performed on basalt, gravel, and gneiss/schist. The shear strength of the basalt rockfill samples was consistently higher across all tested void ratios, with a clear trend of lower strengths at higher void ratios. Similar analyses were conducted for all rock types contained in the database, and each was sorted into one of three categories based on its measured strength: good, moderate, or low quality. A summary of the rock types assigned to each quality category is presented in Table 5.

Fig. 10 shows a scatterplot of $\varnothing'_1$ and e categorized by assigned rockfill quality, along with linear models predicting $\varnothing'_1$ as a

Table 3. Average relative error introduced with the linear approximation method for $\varnothing'_1 = 41°$

					True values		Linear approximation		
m	σ_3 (MPa)	σ_1 (MPa)	$\varnothing'_{PS}$ (degrees)	φ (degrees)	σ_n (MPa)	τ (MPa)	σ_n (MPa)	τ (MPa)	Avg. relative error (%)
1	0.302	1.454	41.0	41.0	0.500	0.435	0.500	0.435	—
0.9	0.283	1.501	43.1	40.0	0.500	0.466	0.476	0.445	5
0.8	0.260	1.542	45.4	38.7	0.500	0.501	0.444	0.450	11
0.7	0.232	1.576	48.0	36.9	0.500	0.537	0.404	0.449	18
0.6	0.198	1.600	51.3	34.7	0.500	0.576	0.352	0.439	27
0.5	0.155	1.610	55.5	31.7	0.500	0.619	0.283	0.412	38

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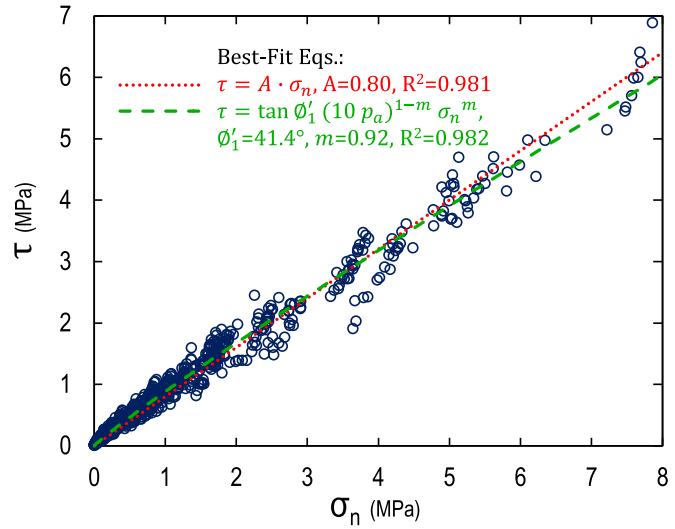


Fig. 4. (Color) Best-fit linear and power-law models for the full testing database (938 tests).

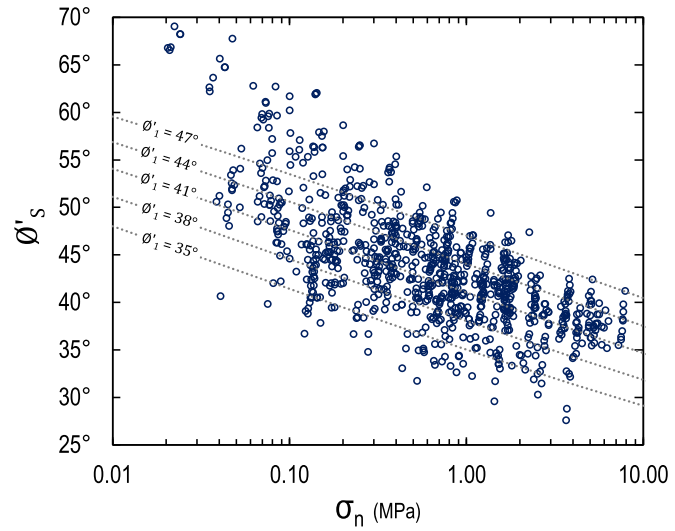


Fig. 5. (Color) Normal stress versus the measured secant friction angle for all tests in the database (938 tests).

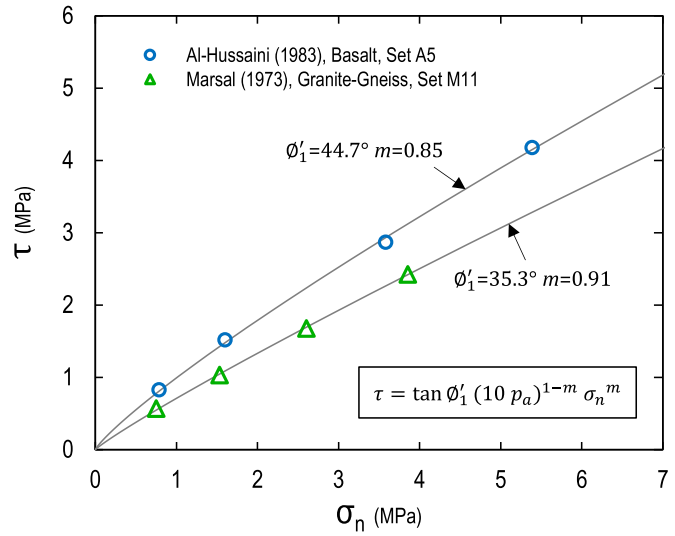


Fig. 6. (Color) Best-fit parameters for two triaxial testing sets.

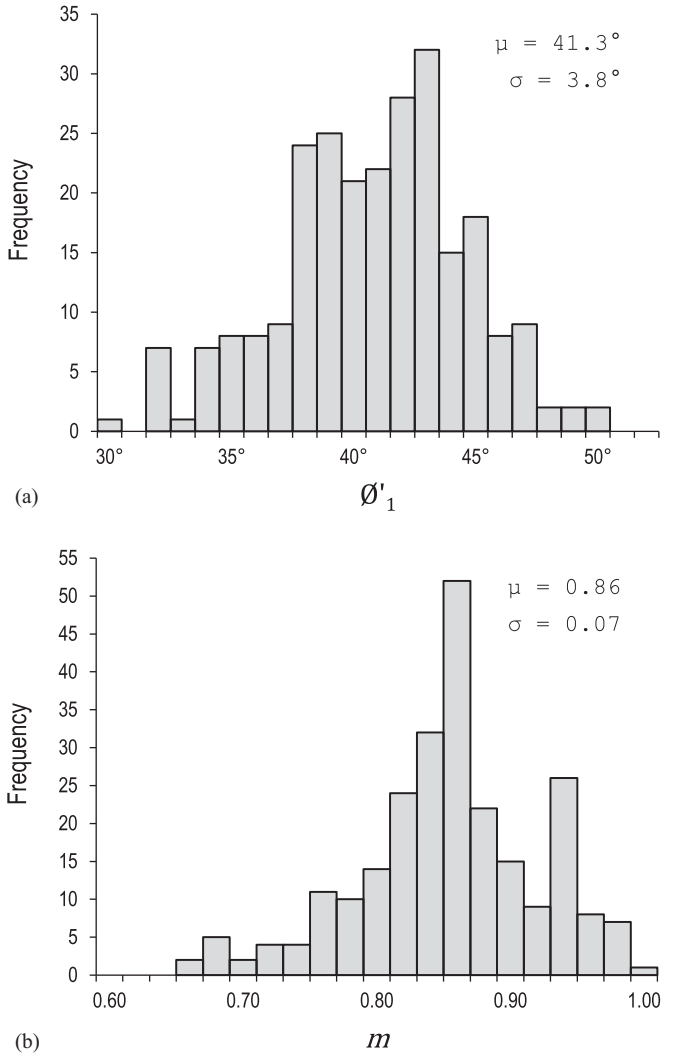


Fig. 7. Measured values of (a) $\emptyset'_1$ parameter; and (b) m parameter for all test sets in the database (249 sets total).

Table 4. Correlation coefficients between measured strength values and laboratory sample parameters

Parameter	E	%Gravel	D_{50}	C_u	C_c	UCS	Ang
$(\tau/\sigma_n)_f$	0.16	0.17	0.05	0.03	–0.02	–0.04	0.06
$\emptyset'_1$	–0.34	0.30	–0.19	–0.02	0.21	0.18	0.08
m	–0.03	–0.32	–0.24	0.02	0.06	0.07	0.06

function of the void ratio. Including rock type improves the model fit, yielding R^2 values of 0.64, 0.36, and 0.26, for good, moderate, and low quality, respectively. While higher R^2 values are preferable, empirical geotechnical models often fall within similar ranges due to the inherent variability of natural materials and diverse sources within a large data set. The quantified model is summarized in Eq. (25), where the secant friction angle intercept at a theoretical void ratio of zero is denoted as $\emptyset'_{e=0}$, and ω represents the slope on a plot of e versus $\emptyset'_1$, as shown in Fig. 10. A value of 20° is proposed to scale the secant friction angle with respect to void ratio, consistent with previous studies that demonstrated values ranging from 15° to 25° (Marachi et al. 1969; Nakayama et al. 1982;

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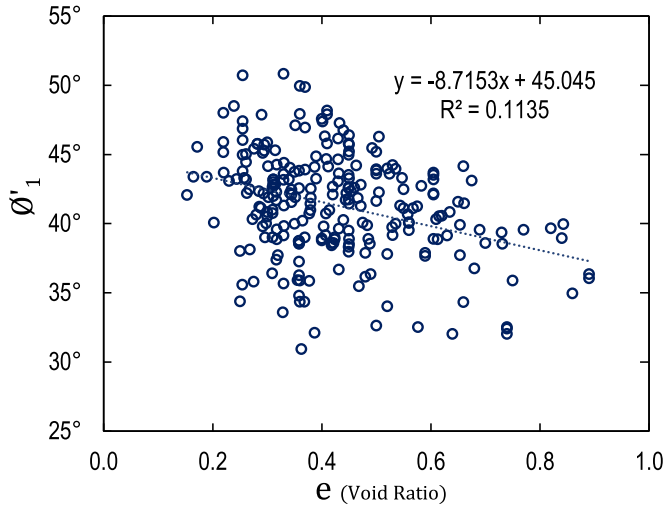


Fig. 8. (Color) Measured $\varnothing'_1$ versus void ratio for all test sets.

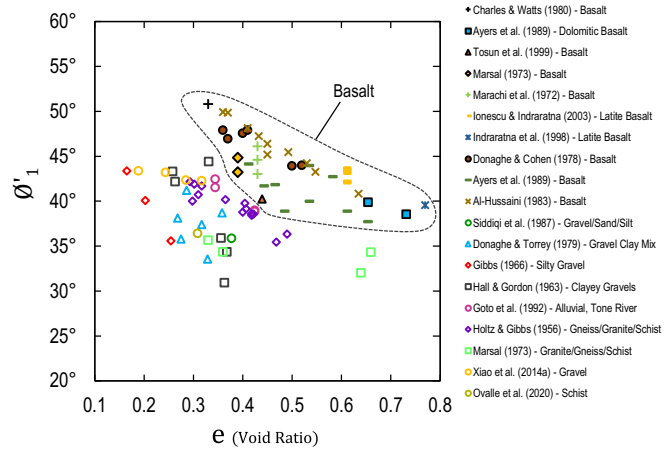


Fig. 9. (Color) Measured $\varnothing'_1$ for test sets on basalt, gravel, and gneiss/schist.

Douglas 2002)

$$\varnothing'_1 = \varnothing'_{e=0} - \omega \cdot e \approx \varnothing'_{e=0} - 20^\circ \cdot e \tag{25}$$

$$\text{For good} - \text{quality rockfill}, \quad \varnothing'_{e=0} = 53^\circ$$
$$\text{For moderate} - \text{quality rockfill}, \quad \varnothing'_{e=0} = 50^\circ$$
$$\text{For low} - \text{quality rockfill}, \quad \varnothing'_{e=0} = 47^\circ$$

The general trends in Table 5 indicate that materials with higher Hoek–Brown m_i values (Hoek and Brown 2019) are typically classified as good or moderate quality, whereas those with lower m_i values tend to fall into the low-quality category. Although particle angularity ratings in the database did not strongly correlate with strength (Table 4), rock types expected to have rounded particles —such as those with a fluvial origin—are generally classified as low quality. For instance, fluvial gravel is categorized as low quality, whereas crushed gravel has a measured strength that is consistent with the moderate-quality category.

Simplified Model Procedure

It is important for a practical model to be useable for any project, even when limited testing data are available. While rock type can generally be identified easily in the field, obtaining the in situ void ratio may not always be feasible. To enable rapid estimation of the shear strength for a wide range of projects, the proposed model was simplified into three levels of compaction—dense, medium, and loose—as outlined in Table 6. These categories are based on typical void ratio ranges for rockfill (Dawson et al. 1998; Das 2019). The nine combinations of rockfill quality and

Table 5. Rock types assigned to each rockfill quality category

Rock quality	Rock types
Good	Basalt[23]
	Latite basalt[23]
	Dolomitic basalt[2]
	Granite[2]
	Quarried calcareous[3]
	Colluvial banded iron formation[2]
Moderate	Phyllite[23]
	Slate[3]
	Andesite[2]
	Argillite[23]
	Quartzite[23]
	Crushed gravel[2]
	Diorite[3]
	Conglomerate[3]
	Silicified conglomerate[3]
	Amphibolite[1]
	Granitic gneiss[3]
	Rhyolite[3]
Low	Gravel[1]
	Clayey gravel[1]
	Gravelly sand[1]
	Gneiss[3]
	Limestone[23]
	Sandstone[3]
	Siltstone[3]
	Schist[3]
	Shale[3]
	Greywacke[3]
	Tuff
	Porphyritic quartz monzonite[3]
	Argillaceous chert

Note: Origin of rockfill (if reported): [1] = Fluvial; [2] = Crushed; and [3] = Blasted.

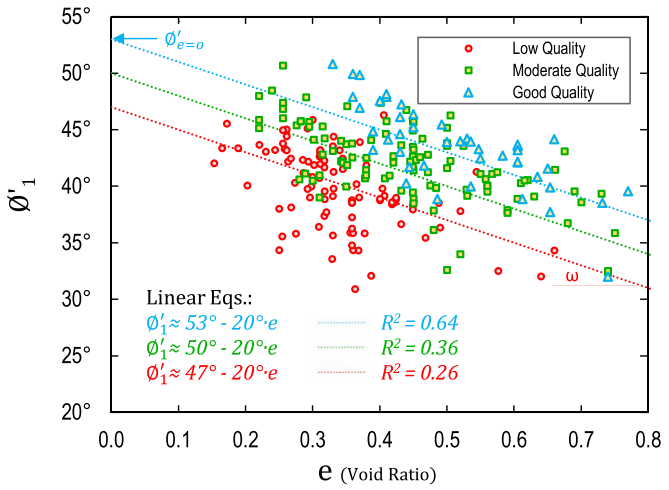


Fig. 10. (Color) Measured $\varnothing'_1$ for test sets sorted by rockfill quality.

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Table 6. Predicted secant friction angle ($\varnothing'_1$) with the proposed model for different rockfill qualities and levels of compaction

	Compaction		
Rockfill quality	Dense $e = 0.3$	Medium $e = 0.45$	Loose $e = 0.6$
Good	47°	44°	41°
Moderate	44°	41°	38°
Low	41°	38°	35°

Table 7. Customized model parameters for three rock types in the database

Rock type	No. of tests	No. of studies	ω (degrees)	$\varnothing'_{e=0}$ (degrees)	m	R^2
Basalt	176	9	15	52	0.85	0.997
Quartzite	55	4	14	50	0.89	0.998
Gravel	85	8	17	46	0.91	0.995

compaction level result in five unique strength envelopes when Eq. (25) is applied: 47°, 44°, 41°, 38°, and 35°.

Customized Model Procedure

Eq. (25) was merged into Eq. (8) to further generalize the strength criterion, as shown in Eq. (26). With Eq. (26), a customized strength model can be created for any dataset, accounting for the effects of compaction, rock type quality, and envelope curvature. The number of fitting parameters for a model depends on the available test data:

1. For a single test, $\varnothing'_1$ is the only fitting parameter using Eq. (9).
2. For a set of tests at different confining pressures, $\varnothing'_1$ and m are the fitting parameters using Eq. (8).
3. For a set of tests at different confining pressures and void ratios, $\varnothing'_{e=0}$, m, and ω are the fitting parameters using Eq. (26).

Three examples of customized strength models for individual rock types from the database analyzed in this study are given in Table 7. By considering only a single rock type (e.g., basalt, quartzite, or gravel), improvements in model fitting accuracy were achieved

$$\tau = \tan(\varnothing'_{e=0} - \omega \cdot e)(10p_a)^{1-m}\,\sigma_n^m \tag{26}$$

Model Performance

Using Eqs. (9) and (25), and Table 5, the predicted shear strength was calculated for every test in the database. The measured versus predicted shear strength for all tests are shown in Fig. 11, with the model yielding an R^2 value of 0.991.

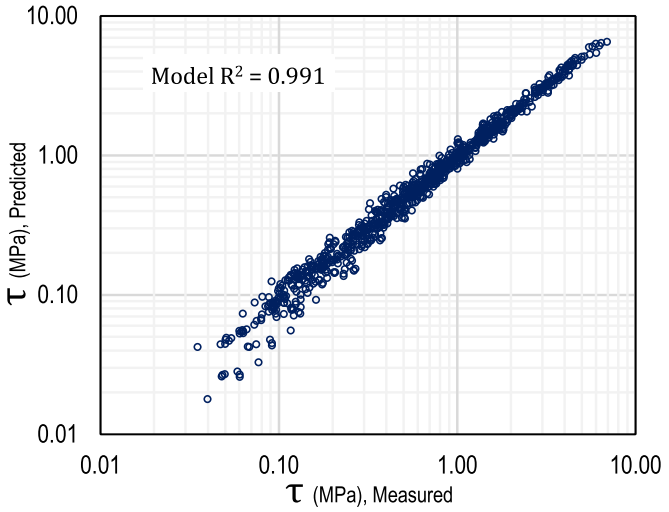


Fig. 11. (Color) Measured shear strength versus predicted shear strength with the proposed model.

Comparison with Published Models

In addition to the proposed model, shear strength was estimated for the tests in the database using four other empirical rockfill strength models for comparison: those of Barton and Kjærnsli (1981), Douglas (2002), Duncan et al. (2014), and Andjelkovic et al. (2018). Readers interested in the details of these models are referred to the original publications. The R^2 values for each model are summarized in Table 8, along with the R^2 values for the simple linear and power-law models for reference. These simple models, illustrated in Fig. 4, serve as baseline fitting targets, because they only require normal stress as the input.

The Andjelkovic et al. (2018) model requires three input parameters and had the poorest performance, with an R^2 of 0.534. The Barton and Kjærnsli (1981) model, which requires six input parameters, including the basic friction angle ($\varnothing_{basic}$), achieved a maximum R^2 of 0.985 when $\varnothing_{basic}$ was set to 31°. Because $\varnothing_{basic}$ was not reported for most tests in the database, it was treated as a fitting parameter. The Duncan et al. (2014) model is based on relative density, which can be calculated directly if the minimum (e_{min}) and maximum (e_{max}) void ratios are known. When these values were unavailable, the relations proposed by Youd (1973) were utilized for estimation. This resulted in an R^2 of 0.984 for all samples and an improved R^2 of 0.986 when considering only samples with $C_u > 4$, as suggested by Duncan et al. (2014). The Douglas (2002) model performed the best among the previously published models, with an R^2 of 0.987, although it has a complex formulation and requires several input parameters. In contrast, the proposed model resulted in the highest R^2 value in addition to being the simplest.

Table 8. Comparison of seven rockfill shear strength models applied to the database

Model	Required input parameters	Training data (No. tests)	R^2
Andjelkovic et al. (2018)	σ_n, e, UCS	118	0.534
Linear regression	σ_n	938	0.981
Power regression	σ_n	938	0.982
Barton and Kjærnsli (1981)	σ_n, e, UCS, D_{50}, Angularity, $\varnothing_{basic}$	Not disclosed	0.985
Duncan et al. (2014)	σ_n, e, e_{min}, e_{max}	226	0.986
Douglas (2002)	σ_n, e, UCS, D_{max}, Angularity, %Fines	869	0.987
Proposed model	σ_n, e, Rock Type	938	0.991

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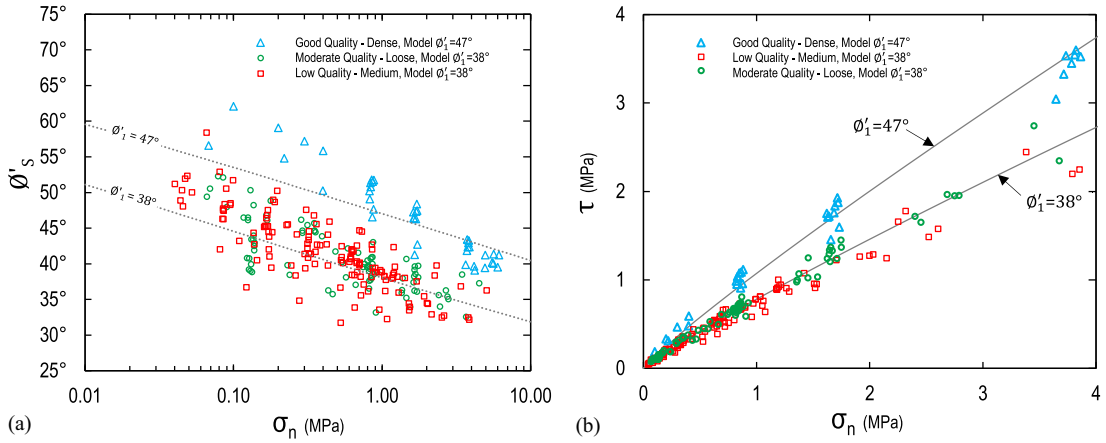


Fig. 12. (Color) Normalized (a) secant friction angle; and (b) shear strength data for good-quality rockfill with $e < 0.425$, moderate-quality rockfill with $e \geq 0.525$, and low-quality rockfill with $0.325 \leq e < 0.45$.

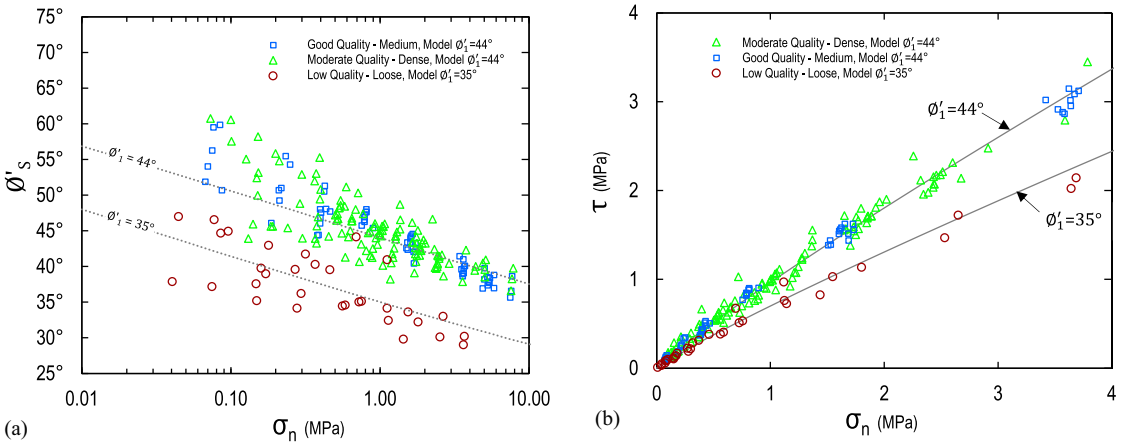


Fig. 13. (Color) Normalized (a) secant friction angle; and (b) shear strength data for good-quality rockfill with $0.425 \leq e < 0.575$, moderate-quality rockfill with $e < 0.375$, and low-quality rockfill with $e \geq 0.45$.

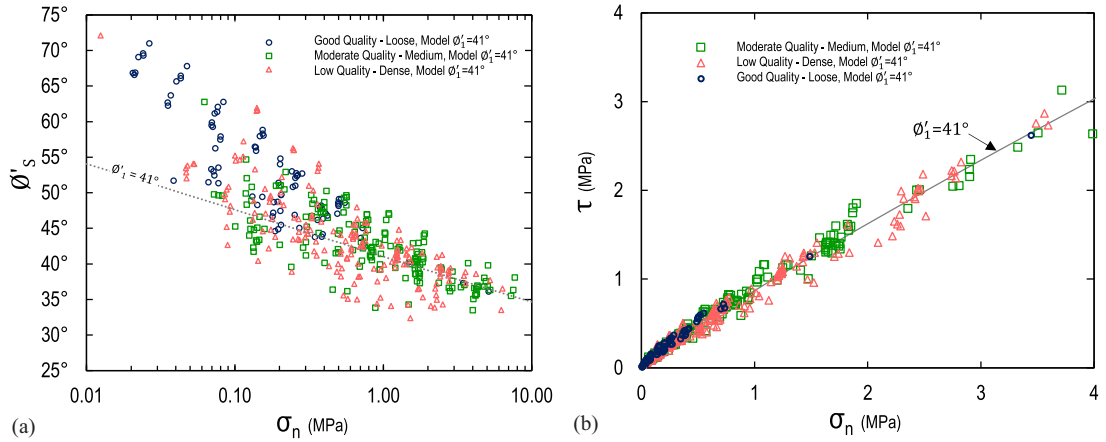


Fig. 14. (Color) Normalized (a) secant friction angle; and (b) shear strength data for good-quality rockfill with $e \geq 0.575$, moderate-quality rockfill with $0.375 \leq e < 0.525$, and low-quality rockfill with $e < 0.325$.

Visual Evaluation of Model Accuracy

Each test in the database corresponds to a unique rock type and void ratio, meaning that the results between tests are not directly comparable without accounting for how these factors influence strength. For example, if the target void ratio (e_{target}) for comparison is $e_{target} = 0.3$, but one of the tests was conducted at $e = 0.4$, an adjustment is needed to make the strength comparable at the target void ratio. The proposed shear strength model was applied to normalize each test result to a desired e_{target}, as outlined in

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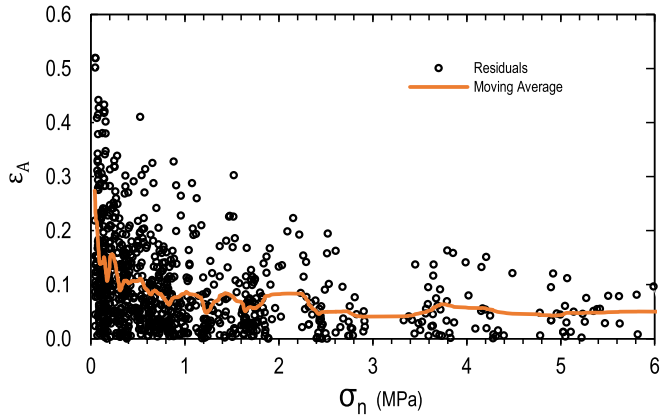


Fig. 15. (Color) Relative absolute error by normal stress.

Eqs. (27)–(29), where $\varnothing'_{adj}$ represents the adjustment to the modeled friction angle, τ_{cor} is the shear strength correction value, and τ_{adj} is the adjusted shear strength

$$\varnothing'_{adj} = -20^{\circ} \cdot (e_{target} - e) \tag{27}$$

$$\tau_{cor} = [\tan(\varnothing'_1 + \varnothing'_{adj}) - \tan\varnothing'_1](10p_a)^{0.1}\sigma_n^{0.9} \tag{28}$$

$$\tau_{adj} = \tau + \tau_{cor} \tag{29}$$

To provide a visual representation of the model accuracy, the relationship between the normal stress and normalized secant friction angle for each test in the database is displayed in Figs. 12(a), 13(a), and 14(a). The data have been organized into the nine categories listed in Table 6. The e_{target} applied for each category follows the values in Table 6: 0.3 for dense, 0.45 for medium, and 0.6 for loose. These figures also include the predicted secant friction angles from the model for the five unique strength envelopes. Figs. 12(b), 13(b), and 14(b) present a comparison between normal stress and normalized shear strength, overlaying the predicted strength envelopes from the proposed model for each category. The clear organization of test results by void ratio and rock type quality highlights the model's capability to characterize material behavior data across varying densities and stress levels.

Model Accuracy by Stress Range

The relative absolute residual error (ε_A) was calculated for each test in the database using Eq. (30) and is plotted against the normal stress in Fig. 15. The plot reveals that model predictions of strength exhibit significantly greater residual error at low values of σ_n (i.e., tests at low σ_n show greater scatter, even after adjusting for rock type and void ratio). The error was quantified in practical terms by solving Eq. (9) for $\varnothing'_1$ and calculating the measured secant friction angle ($\varnothing'_{1-test}$) for each test, as shown in Eq. (31). The relative absolute residual error in the estimation of $\varnothing'_1(\varepsilon_{A-\varnothing'_1})$ was then computed with Eq. (32). Table 9 summarizes the estimation error

Table 9. Measured error in $\varnothing'_1$ by stress range (assuming $m = 0.9$)

Stress range (MPa)	Mean absolute error (degrees)	Standard deviation of absolute error (degrees)
$\sigma_n < 0.1$	6.8	±6.6
$0.1 \le \sigma_n < 0.8$	3.4	±2.8
$\sigma_n \ge 0.8$	1.7	±1.5

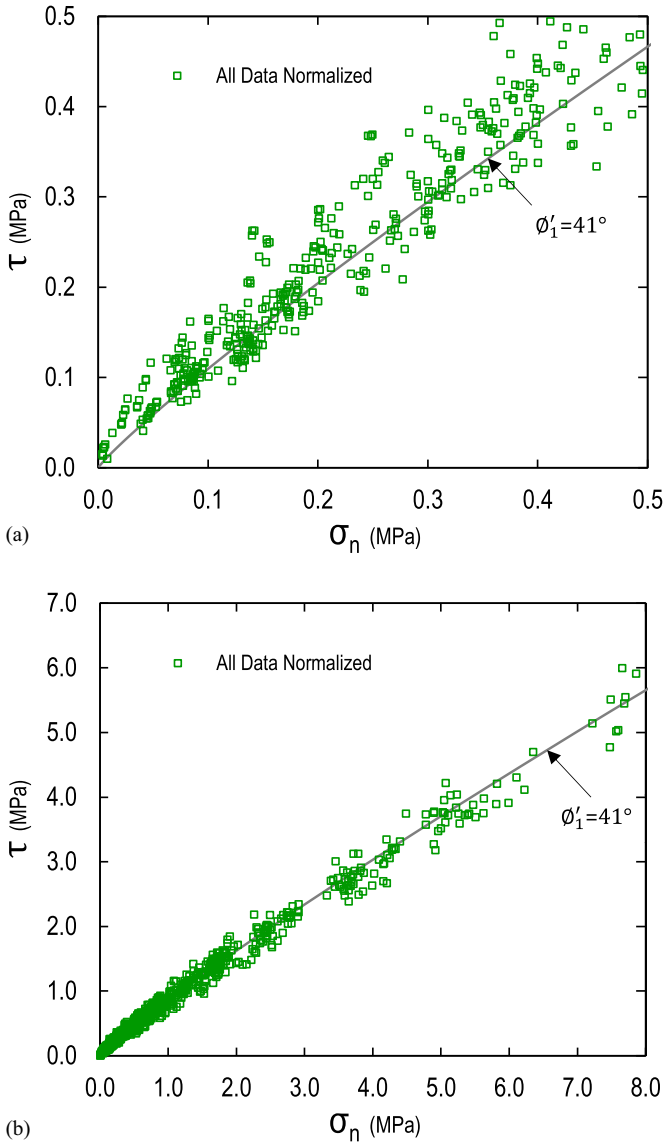


Fig. 16. (Color) Measured strength of all tests normalized to $e_{target} = 0.45$ and moderate-quality rock type in the (a) low stress range; and (b) high stress range.

of $\varnothing'_1$ by stress range for the proposed model, showing that the mean absolute error is approximately 7° for $\sigma_n < 0.1$ MPa and decreases to <2° for $\sigma_n > 0.8$ MPa.

The degree of scatter in the data across different stress ranges was visualized by normalizing all tests in the database to the same void ratio and rock type quality, as shown in Figs. 16(a and b). At $\sigma_n <$ 0.1MPa, a systematic underestimation of shear strength is observed relative to the normalized strength predicted by the model for moderate-quality rockfill with medium compaction ($\varnothing'_1 = 41^{\circ}$)

$$\varepsilon_A = \frac{|\tau_{pred} - \tau_{test}|}{\tau_{test}} \tag{30}$$

$$\varnothing'_{1-test} = \text{atan}\left(\frac{\tau}{(10\,p_a)^{0.1}\,\sigma_n^{0.9}}\right) \tag{31}$$

$$\varepsilon_{A-\varnothing'_1} = \frac{|\varnothing'_{1-pred} - \varnothing'_{1-test}|}{\varnothing'_{1-test}} \tag{32}$$

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Table 10. Summary of the 60-600 Guideline for fitting a power-law strength model for rockfill

Guidance	Triaxial σ_3 (kPa)	Direct shear σ_n (kPa)
Do not use tests below this stress level to calibrate the failure criterion	60	100
Try to incorporate at least one test above this stress level	600	1,000

60-600 Guideline

The evaluation of model accuracy across different stress ranges revealed increased scatter and an underestimation of strength at very low normal stresses ($\sigma_n < 100$ kPa). This finding indicates that if tests at low confinements are used to calibrate a power-law strength criterion, shear strengths could be significantly overestimated for $\sigma_n > 100$ kPa. Therefore, new guidance is proposed termed the *60-600 Guideline* for power-law strength models of rockfill. Specifically, when fitting a power-law model, triaxial tests with $\sigma_3 <$ 60 kPa should not be used ($\sigma_n < 100$ kPa for direct shear tests), and at least one test in each set should include a confining stress of $\sigma_3 > 600$ kPa ($\sigma_n > 1{,}000$ kPa for direct shear tests) to result in reliable predictions of strength. This guideline is summarized in Table 10. It is important to note that while the guideline will result in more accurate estimates of strength for $\sigma_n > 100$ kPa, it may produce conservative estimates of strength for $\sigma_n < 100$ kPa.

Model Accuracy by Sample Parameter

The relative residual error (ε) for each test in the database was calculated using Eq. (33) and is plotted against various parameters in Figs. 17(a–d). A residual error of zero indicates an exact prediction of strength for that test, a negative value indicates an underestimation of strength, and a positive value indicates overestimation. Additional relative residual error plots for other parameters in the database can be found in Figs. S1(a–o)

$$\varepsilon = \frac{\tau_{\text{pred}} - \tau_{\text{test}}}{\tau_{\text{test}}} \tag{33}$$

Key observations from the error plots include the following:

1. The model shows no clear trend in ε with respect to UCS, $D_{\max}$, $D_S/D_{\max}$, or C_C. However, this does not necessarily imply that these parameters have no influence on the shear strength of rockfill. Their effects may be indirectly accounted for by the consideration of rock type and void ratio. Alternatively, their influence might be too minor to be observable above the noise in the data set, given the diversity of samples analyzed. Previous studies have suggested that these parameters can affect the shear strength of rockfill. For instance, Frossard et al. (2012) and Ovalle et al. (2014) observed a decrease in shear strength with increasing $D_{\max}$.
2. The model does not exhibit a trend in the mean value of ε with respect to %Gravel or %Sand. However, the standard deviation of the error increases for samples with gravel content below 50% or sand content above 50%.
3. The model tends to overpredict shear strength when %Fines exceeds 30%. The standard deviation of the error increases for samples exceeding 10% fines content.
4. The model tends to underpredict the shear strength at very low normal stresses ($\sigma_n < 100$ kPa).
5. The model overpredicts shear strength for weak samples ($\varnothing'_1 < 35°$).
6. The model underpredicts shear strength for sample sets with a highly curved failure envelope ($m < 0.8$).

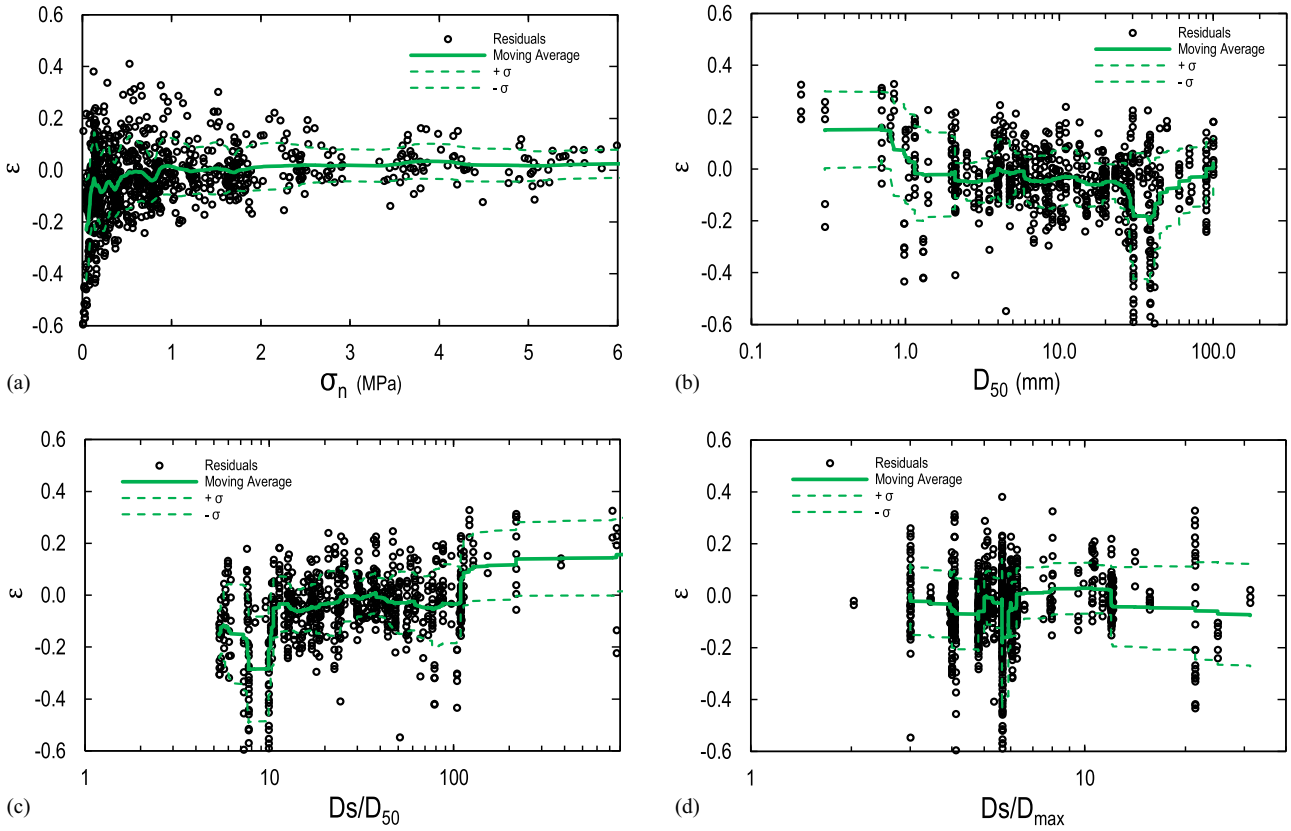


Fig. 17. (Color) Relative residual error versus the parameters of (a) σ_n; (b) D_{50}; (c) D_S/D_{50}; and (d) D_S/D_{Max}.

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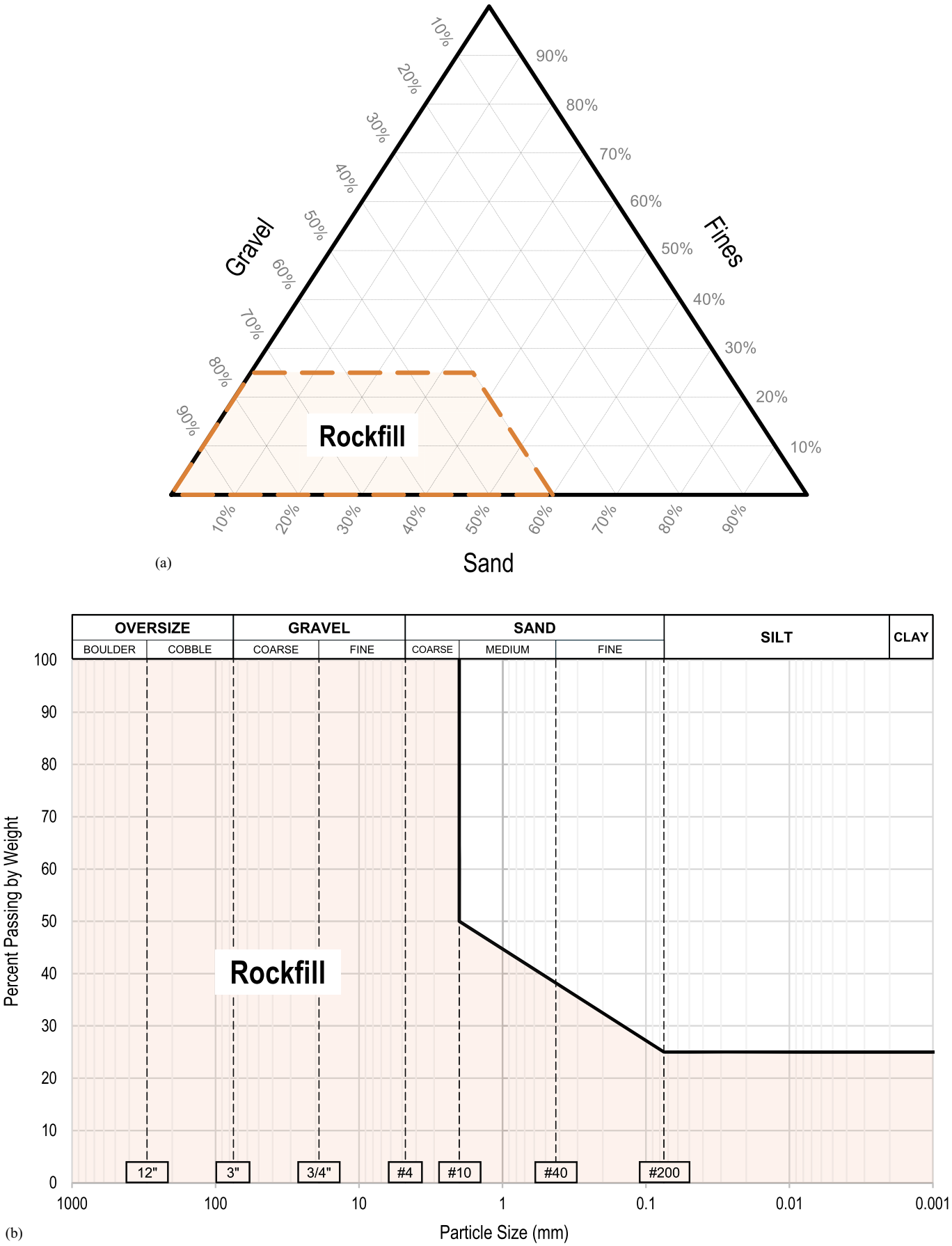


Fig. 18. (Color) Proposed application range for the rockfill shear strength model on the (a) ternary soil plot; and (b) cumulative particle size plot.

7. The model underpredicts shear strength when the median particle size is large relative to the triaxial cell size ($D_S/D_{50} < 10$). In contrast, D_S/D_{max} shows no trend in residual error for any range of tested values. This suggests that D_{50} may be a better predictor of the representative elementary volume (REV) for strength testing on rockfill. This finding runs counter to the current guidelines for testing coarse-grained soils, which use D_{max} to determine the REV (D3080/D3080M-23, ASTM 2023b).
8. The model tends to underpredict shear strength for poorly graded (uniform) rockfill ($C_u < 2$) and for very loose samples ($e > 0.70$). There is considerable overlap between these two populations, because soils with low C_u generally exhibit higher void ratios, even when compacted (Youd 1973). The issue is not that these samples are inherently stronger than average, but rather that the model overcorrects for their high void ratios, leading to an underestimation of strength.
9. The model overpredicts shear strength for samples with $D_{50} <$ 2mm and when the particle size is small relative to the triaxial cell ($D_S/D_{50} > 100$). There is significant overlap between these two groups, and it could be argued that particles of this size should not be classified as rockfill.

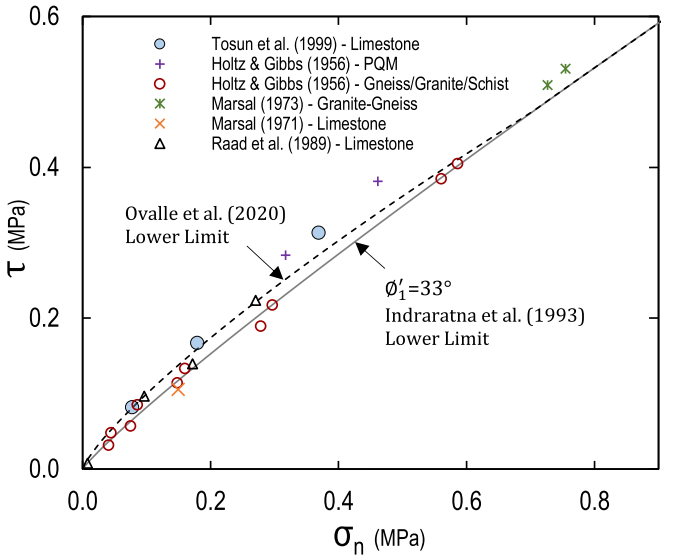


Fig. 19. (Color) Tests on low-quality rockfill types with $e > 0.45$ normalized to $e_{target} = 0.6$.

Discussion

Range of Model Application (Definition of Rockfill)

Rockfill has traditionally been defined as soil with $D_{50} > 4.75$ mm (Xiao et al. 2014a). To better delineate the range of applicability of the proposed rockfill strength model, the residual error plots in Figs. S1(a–o) were examined in conjunction with published soil mixing tests. Important observations include the following:

1. The mean residual model error is near zero for samples with $D_{50} > 2$mm.
2. The standard deviation of the residual error increases for samples with %Gravel < 50% or %Sand > 50%.
3. The model tends to overpredict shear strength when %Fines > 30%, and the standard deviation of the residual error increases for samples with %Fines > 10%.
4. Based on a compilation of published soil mixing tests, Park and Santamarina (2017) observed that the transition in mechanical behavior from gravel to sand occurs between 47% and 70% sand content, and from coarse- to fine-grained soil between 10% and 42% fines content.

Based on these findings, the proposed application range for the strength model—in which soil is expected to exhibit the mechanical behavior of rockfill—is for soils with %Gravel > 40% and %Fines < 25%. This range is illustrated on a ternary textural chart [Fig. 18(a)] and a particle size plot [Fig. 18(b)]. If a soil's gradation curve falls entirely within the colored zone, the rockfill strength model is anticipated to be applicable. Fig. 18(b) suggests a simple screening test: if < 50% passes the No. 10 sieve (2 mm) and <25% passes the No. 200 sieve (0.075 mm), a soil is likely to exhibit rockfill mechanical behavior.

Lower Bound of Rockfill Shear Strength

Indraratna et al. (1993) and Ovalle et al. (2020) each proposed a lower limit strength envelope for rockfill. The details of these strength envelopes can be found in the original publications. The Ovalle et al. envelope is stronger for $\sigma_n < 0.7$ MPa, and the two envelopes converge and become identical for $\sigma_n > 0.7$ MPa. To visually evaluate the performance of the models, the low-quality rockfill types tested in a loose state were normalized to $e_{target} =$ 0.6 for comparison, as shown in Fig. 19. These tests theoretically represent the lower limit of rockfill shear strength in the database. Fig. 19 demonstrates that the model of Indraratna et al. (1993) better captures the lower bound of the data, which is reproduced using Eq. (9) with the assumption $\varnothing'_1 = 33^\circ$.

Limitations

The database analyzed spans a broad range of rock types. However, it is important to note the uneven representation of rock types in the database. For instance, there are 55 tests on quartzite, compared with only 13 tests on diorite. The reliability of strength predictions may be reduced when using Table 5 for less-represented materials. Furthermore, weathering and alteration can degrade rockfill strength (Sayão et al. 2005), meaning that rock type may not always be a consistent indicator of strength. The lowest UCS value in the database is 25 MPa, meaning that rockfills composed of weaker particles could exhibit lower strengths than predicted by the empirical model. Additionally, the database does not contain every potential variation or alteration of the included rock types.

Testing methodologies can also introduce limitations. The effects of particle scalping or downsizing—particularly for coarse rockfill with substantial large-diameter fractions—may impact the representativeness of the database for certain applications. Moreover, partially saturated or undrained conditions were not explicitly addressed in this study, meaning that scenarios involving pore pressure may require more advanced analyses.

While the database and proposed models offer guidance, they may not fully capture all critical factors in every design scenario. Consequently, site-specific strength testing is strongly recommended, particularly for high-consequence projects.

Conclusions

A novel shear strength criterion and empirical model for rockfill has been developed based on a database of over 900 large-scale triaxial tests. The main conclusions are summarized as follows:

1. Shear strength criterion: A strength criterion was introduced that includes a single fitting parameter defined as the secant friction

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angle at a normal stress of 10 atmospheres (approximately 1 MPa). By using a physically meaningful fitting parameter expressed in degrees, the model reduces the risk of errors and circumvents unit conversion complications typically associated with power-law models. The strength criterion can replicate many previously proposed rockfill strength envelopes, with the added benefits that the strength profile transitions smoothly between envelopes and the criterion can be applied to fit any data set expressed in either shear/normal or principal stresses.

2. Empirical model: An empirical model that employs a simple equation to estimate the fitting parameter for the rockfill failure criterion based on void ratio and rock type was introduced. This model enables a rapid estimation of rockfill strength using readily measured field properties, making it a practical tool for project planning and on-site applications.
3. Definition of rockfill: An analysis of the testing database and model accuracy indicated that soils with %Gravel > 40% and %Fines < 25% are likely to exhibit the mechanical behavior of rockfill.
4. Rockfill testing guideline: The evaluation of model accuracy revealed that calibrating a nonlinear power-law strength criterion with tests at very low confinements may lead to an over- or underestimation of shear strength for $\sigma_n > 100$ kPa. Based on this, the 60-600 Guideline for power-law strength models of rockfill was proposed, recommending the avoidance of triaxial confining stresses below 60 kPa and ensuring that at least one test in each set includes a confining stress above 600 kPa to obtain reliable strength predictions.

The proposed models bridge the gaps between laboratory testing, modeling, and field application, offering accurate predictions for a wide range of rockfill types and stress conditions. Their simplicity and ease of calibration make them adaptable, even when data are limited. By providing a practical approach to rockfill strength estimation, these models may contribute to safer and more efficient design of rockfill structures.

Data Availability Statement

The rockfill test database supporting this paper is available at https://doi.org/10.5281/zenodo.14665504.

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Notation

The following symbols are used in this paper:

Ang = particle angularity rating;
C_c = coefficient of curvature;
C_u = coefficient of uniformity;
D_{max} = maximum particle diameter;
D_S = test diameter;
D_{50} = median particle diameter;
e = void ratio at start of test;
e_{max} = maximum void ratio;
e_{min} = minimum void ratio;
e_{target} = target void ratio;
K = strength coefficient;
m = curvature coefficient;
m_i = Hoek–Brown criterion fitting parameter;
p_a = standard atmospheric pressure;
R = correlation coefficient;
R^2 = coefficient of determination;
β = angle of failure plane with respect to σ_3;
ε = relative residual error;
ε_A = relative absolute residual error;
$\varepsilon_{A-\varnothing_1''}$ = relative absolute residual error in the estimation of $\varnothing_1'$;
μ = mean;
σ = standard deviation;
σ_n = normal effective stress at failure;
σ_1 = major principal effective stress at failure;
σ_3 = minor principal effective stress at failure;
τ = shear stress at failure;
τ_{adj} = adjusted shear strength;
τ_{cor} = shear strength correction value;
φ = instantaneous friction angle;
$\varnothing_{basic}$ = basic friction angle;
$\varnothing_{adj}'$ = adjustment to the modeled friction angle;
$\varnothing_{e=0}'$ = secant friction angle intercept at a theoretical void ratio of zero;
$\varnothing_{PS}'$ = secant friction angle in terms of σ_1 and σ_3, peak-state friction angle;
$\varnothing_S'$ = secant friction angle in terms of σ_n and τ;
$\varnothing_1'$ = secant friction angle at $\sigma_n = 10\ p_a$;
$\varnothing_{1-test}'$ = tested value of $\varnothing_1'$;
ω = slope parameter on a plot of e versus $\varnothing_1'$;
%Gravel = gravel content;
%Sand = sand content; and
%Fines = fines content.

Supplemental Materials

Fig. S1 is available online in the ASCE Library (www.ascelibrary.org).

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